Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Normal Distribution

SOLUTIONS

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1. $Z = \frac{54-50}{4} = 1$; $\Phi(1) \approx 0.8413$.
2. $Z = \frac{108-100}{5} = 1.6$; $1 - \Phi(1.6) \approx 0.0548$.
3. $Z = \pm 1$; $\Phi(1) - \Phi(-1) \approx 0.6826$.
4. $\Phi(1.5) \approx 0.9332$. **5.** $2\Phi(1.96) - 1 \approx 0.95$.
6. $\frac{10-\mu}{2} = z_{0.3} \approx -0.524 \Rightarrow \mu \approx 11.05$.
7. $\frac{35-30}{\sigma} = z_{0.8} \approx 0.842 \Rightarrow \sigma \approx 5.94$.
8. Mean 0, variance 1.
9. $z_{0.9} \approx 1.282$; $a = 60 + 1.282 \times 6 \approx 67.7$.
10. $Z = 1$; $1 - \Phi(1) \approx 0.1587$.
11. $Z = 1.5$; $1 - \Phi(1.5) \approx 0.0668$.
12. $Z = -1$ and 0.5; $\Phi(0.5) - \Phi(-1)$.
13. Sketch standard normal; shade right of 1.2.
14. $1 - \Phi\left(\frac{a-\mu}{\sigma}\right)$. **15.** One SD.
16. $c = 1.96 \times 8 \approx 15.68$.
17. $z_{0.9} \approx 1.282$; score $= 70 + 1.282 \times 7 \approx 79.0$.
18. Solve two equations for μ (may be inconsistent — comment).
19. $Z = \pm 0.5$ and 1; evaluate Φ.
20. Q_1, Q_3 at $z \approx \pm 0.6745$; IQR ≈ 5.40.
21. Continuous distribution; probability at a point is zero.
22. $X + Y \sim N(35, 13)$; $P(X + Y > 40) = 1 - \Phi\left(\frac{5}{\sqrt{13}}\right)$.
23. $z \approx 1.96$.
24. Total $\sim N(480, 100)$; $P(T > 490) = 1 - \Phi(1)$.
25. Continuity correction: discrete $X \leq k \approx$ continuous $X < k + 0.5$.